Afternoon Lab

EBIKE101 | CTS Cohort 6 | Day 33 | Monday, August 31, 2026

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Afternoon Lab: E-bike System Walk and Pre-Ride Inspection

Safe micro-mobility starts before the throttle moves.

Lab purpose

Students inspect an e-bike as a complete system and prove that safety comes before powered operation.

Safety boundary

- No riding today unless the instructor explicitly changes the plan after inspection.
- No throttle tests without instructor approval.
- No powered wheel spin without the bike stable, clear, and controlled.
- Battery remains off/disconnected unless the instructor directs otherwise.
- Any student can call stop-work.
- If something is unknown, damaged, loose, hot, swollen, rubbing, or unclear, mark it for instructor review.

Team roles

Role	Job
Inspector 1	Leads mechanical inspection
Inspector 2	Leads electrical inspection
Recorder	Writes pass/fail notes and risk report
Reporter	Shares findings during debrief

Rotate roles if time allows.

Lab A: Mechanical inspection

1. Stabilize the bike.

2. Identify the frame, fork, wheels, tires, brakes, drivetrain, cockpit, seat, pedals, and cranks.

3. Check for obvious looseness, damage, rubbing, missing parts, or unsafe setup.

4. Use the checklist in the student packet.

5. Mark anything uncertain as R for instructor review.

Lab B: E-bike system inspection

1. Identify battery, motor, controller, display, throttle/PAS, wiring, connectors, charger, and cutoffs if installed.

2. Look for damaged insulation, loose connectors, stressed wires, rubbing, pinch points, unsupported cable runs, and questionable battery condition.

3. Do not plug, unplug, charge, or power anything unless directed.

4. Use the checklist in the student packet.

Lab C: System flow map

Draw the basic flow:

Rider input -> controller -> battery power -> motor -> wheel motion

Then add safety checkpoints:

- brakes
- tire condition
- battery mount
- wiring clearance
- rider control
- area control

Lab D: Risk report

Each team reports:

1. One mechanical risk

2. One electrical risk

3. One rider-control risk

4. One thing that must be checked before tomorrow

Instructor scoring

Category	3 - Job-ready	2 - Acceptable	1 - Needs rework	0 - Unsafe/incomplete
Safety	Holds no-power boundary, uses stop-work thinking	Minor reminders	Frequent reminders	Unsafe action or ignores boundary
Inspection	Checks all major systems	Misses one minor item	Misses several items	Skips inspection
Accuracy	Correctly identifies bike/e-bike parts	Mostly correct	Confuses several systems	Cannot identify core systems
Documentation	Clear pass/fail notes and risks	Basic notes	Incomplete notes	No usable notes
Teamwork	Roles used well	Roles mostly used	Roles unclear	Disruptive or unsafe

Finish early extension

Have teams design a controlled ride-test area on paper. Include:

- start line
- stop line
- cone path
- observer zone
- emergency stop signal
- helmet/PPE check

- instructor checkpoint

Lab close

Before leaving:

- Tools returned
- Bike stable and powered off
- Packets collected
- Risks written on the board
- Tomorrow's first check identified